

BUSINESS PROBLEM & KPI DEFINITION

Kenya Mobile Money & Agent Network Intelligence

Context

Kenya's mobile-money agent network has grown rapidly, CBK reported 381,116 active agents in 2024, rising to 473,536 by December 2025, processing roughly KSh 722.5 billion in monthly transaction value. But growth in agent numbers has not translated into growth in agent economics: average annual M-Pesa agent commissions fell to approximately KSh 112,244 for the year ended March 2026. More agents are splitting a commission pool that isn't growing proportionally.

(All figures above are public CBK/industry statistics used only to ground realistic assumptions. All transaction-level data in this project is synthetic and does not represent Safaricom, Airtel, or any bank's actual data.)

1. Business Problem Statement

You are a Data Analyst at a Kenyan financial-services company operating a large mobile-money agent network. Management has observed that the agent network is expanding, but is unsure whether that expansion is translating into proportional revenue growth, and where operational or risk issues may be eroding profitability.

Central question: How profitable is our agent network, where are we performing well, and where are we losing money or facing operational risk?

This project answers that question using a synthetic but business-realistic dataset, a dimensional data model, and a Power BI report built on SQL-backed data and DAX measures.

2. KPI Definitions

Nine KPIs were defined at the outset to directly answer the central question, and later implemented as DAX measures in the Power BI model.

#	KPI	Formula (Conceptual)	Business Question It Answers
1	Total Transaction Value	SUM(TransactionAmount)	How much money is moving through the network?
2	Transaction Count	COUNTROWS(FactTransactions)	How much activity is happening on the platform?
3	Total Revenue	SUM(Commission)	How much gross revenue does the network generate?

4	Total Processing Cost	$\text{SUM}(\text{ProcessingCost})$	What does it cost to operate the network?
5	Net Revenue	$\text{Total Revenue} - \text{Total Processing Cost}$	What's left after costs; the true profit measure
6	Average Transaction Value	$\text{Total Transaction Value} \div \text{Transaction Count}$	What does a typical transaction look like?
7	Revenue per Active Agent	$\text{Total Revenue} \div \text{DISTINCTCOUNT}(\text{AgentID with } \geq 1 \text{ transaction})$	The key KPI; does network growth dilute individual agent earnings?
8	Success Rate	$\text{Successful Transactions} \div \text{Total Transactions}$	How operationally reliable is the network?
9	Net Revenue Margin	$\text{Net Revenue} \div \text{Total Revenue}$	How efficiently does revenue convert to profit?

Why "Revenue per Active Agent" Is the Central KPI

This metric was identified from the outset as the single most important number in the project because it directly tests the real-world tension described in the Context section above: **CBK data shows agent commissions falling even as the agent network grows**. If that same pattern shows up in this dataset, declining revenue per agent as agent count increases, it would validate a market-wide dilution effect. If it doesn't, that's an equally important (and more actionable) finding.

(Spoiler, documented in full in: the synthetic dataset shows revenue per agent remaining stable across county sizes, no strong dilution effect was observed. This is reported honestly rather than forced to match the hypothesis.)

Project Approach

To answer the central question reliably, this project was built in phases:

1. **Data Model** - a dimensional (star schema) database design in MySQL
2. **Reference Data** - Kenyan counties, regions, and provider market share grounded in real CA Kenya / CBK figures
3. **Synthetic Data Generation** - a Python pipeline producing 1,000,000 statistically realistic transactions
4. **SQL Analysis** - 10 exploratory queries testing the KPIs above against the generated data
5. **Power BI Dashboard** - a 2-page interactive report translating the KPIs and SQL findings into visual, filterable insights
6. **Business Insights Write-Up** - a narrative document synthesizing findings into strategic recommendations